CzarIntel Landing Zone — Architecture Overview

> **Audience:** CTOs, security architects, auditors and platform engineers.
> **Scope:** every component in this repository, how the components connect, and why each design decision was made.

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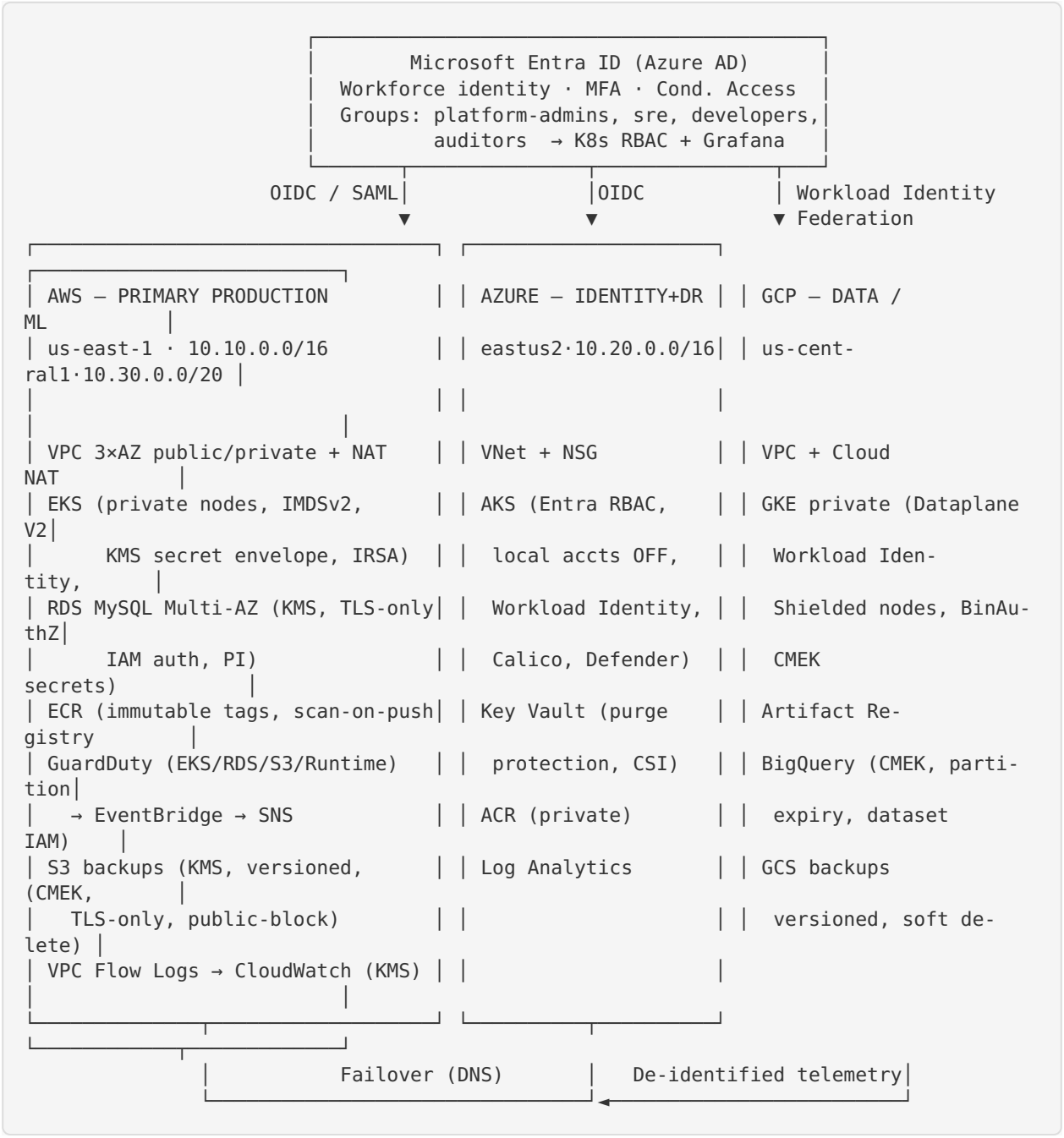
1. Design principles

#	Principle	How it is enforced in this repo
P1	**Zero Trust: never trust, always verify**	Default-deny NetworkPolicies, OIDC/workload identity on every cloud, no static credentials, mTLS-ready service boundaries. See zero-trust.md.
P2	**Everything as code**	Terraform (infra), Helm (apps), ArgoCD (delivery), Ansible (VMs), Gatekeeper (policy), Python-generated Grafana dashboards. Nothing is configured through a console.
P3	**Pull-based delivery**	Jenkins builds, scans and pushes images, then bumps a Git tag. ArgoCD pulls that change into the cluster. CI never holds cluster-admin credentials.
P4	**Right cloud for the job**	AWS runs production, Azure provides identity and DR, GCP runs data/ML. Each cloud is used for what it does best, not duplicated.
P5	**Compliance is a build artefact**	`scripts/compliance_scan.py` runs in CI and fails the build on regressions. Every control is mapped in ../compliance/control-matrix.md.
P6	**Observable by default**	Every workload exposes Prometheus metrics, ships logs to Loki, and is covered by an SLO dashboard and burn-rate alerts.

2. Multi-cloud role allocation



Cloud	Role	Why this cloud
AWS	Primary production: VPC, EKS, RDS, GuardDuty, ECR, S3	Deepest managed-service catalogue and the most mature HIPAA BAA / PCI attestation coverage. GuardDuty adds EKS runtime threat detection.
Azure	Workforce identity (Entra ID) plus warm DR (AKS, Key Vault, ACR)	Most enterprise buyers already run Entra ID. Federating from it avoids a second identity silo, and AKS in east-us2 gives geographic and provider diversity for DR.
GCP	Data and ML: GKE, BigQuery, Artifact Registry	BigQuery is serverless and has column-level governance. GKE Autopilot-grade hardening (Binary Authorization, Dataplane V2) suits batch/ML workloads.

Each cloud is behind its own feature flag (`enable_aws` , `enable_azure` , `enable_gcp`), so a client can adopt one cloud at a time.

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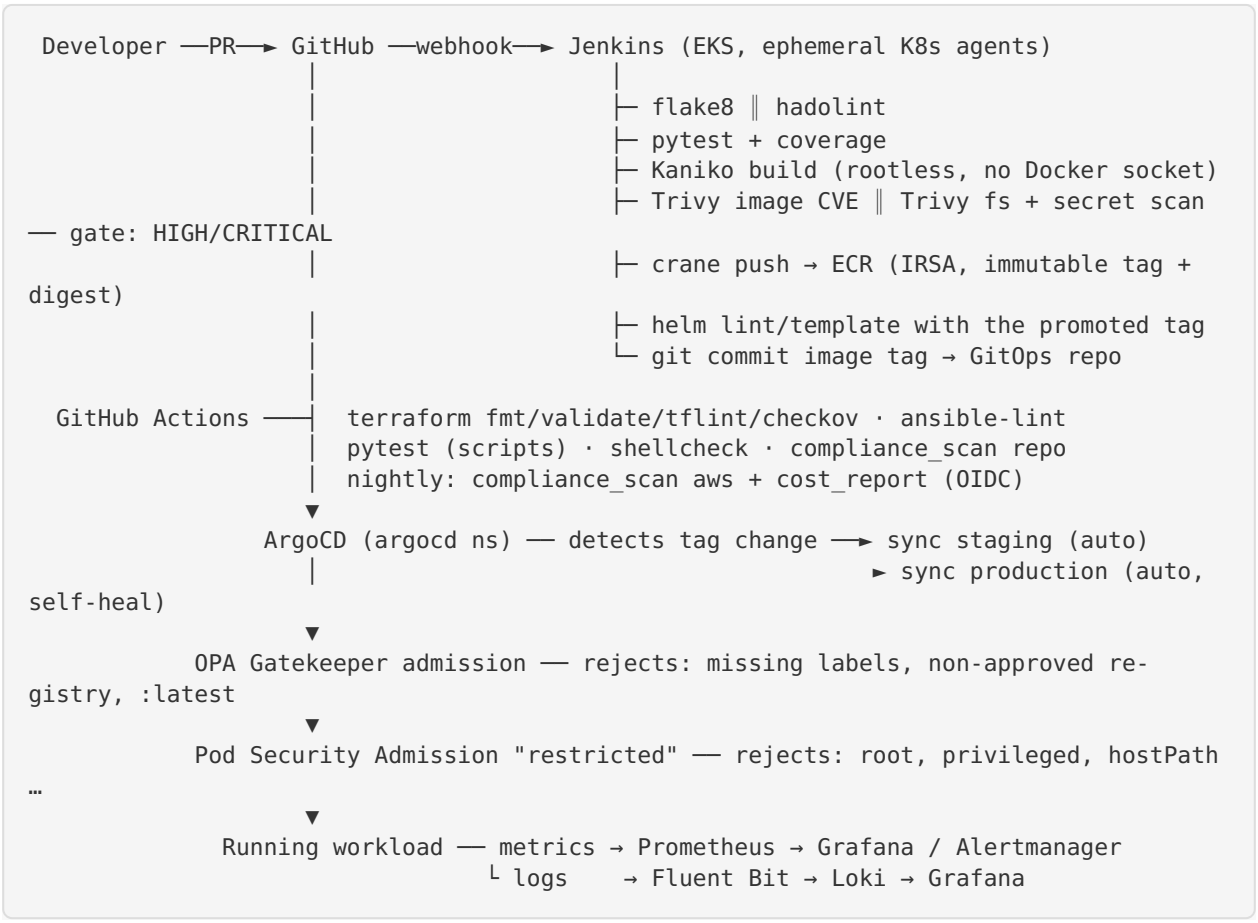
3. Software delivery flow



4. Repository map

Path	Layer	Deep dive
`terraform/`	Multi-cloud infrastructure (root plus `aws`, `azure`, `gcp` modules)	tools/terraform.md, tools/aws.md, tools/azure.md, tools/gcp.md
`docker/`	Hardened api-service (FastAPI) and frontend images	tools/docker.md
`kubernetes/`	Namespaces with PSA, RBAC, NetworkPolicies, Gatekeeper	tools/kubernetes.md
`helm/`	`api-service` and `mysql` charts with per-env values	tools/helm.md
`argocd/`	AppProject plus Applications (GitOps)	tools/argocd.md
`ci-cd/`	Jenkinsfile and Jenkins-on-Kubernetes manifests	tools/jenkins.md
`.github/workflows/`	IaC and platform quality gates	tools/jenkins.md#github-actions-quality-gates
`monitoring/`	kube-prometheus-stack, alert rules, Loki, Grafana dashboards-as-code	tools/observability.md
`ansible/`	CIS hardening, agents, Docker, Jenkins, MySQL/ProxySQL on VMs	tools/ansible.md, tools/mysql.md
`scripts/`	Bootstrap, compliance scanner, cost report, health check	tools/automation-scripts.md
`docs/`	This documentation	—

5. Environments

Environment	Terraform workspace	K8s namespace	Helm values	ArgoCD sync
dev	`dev`	`staging` (shared)	`values.yaml`	manual
staging	`stg`	`staging`	`values-staging.yaml` (2–4 replicas)	automated
production	`prod`	`production`	`values-production.yaml` (4–20 replicas, HPA 65% CPU)	automated plus self-heal

State lives in one S3 bucket (versioned, KMS) with DynamoDB locking. Workspaces keep keys separate under `env:/<workspace>/`.

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6. Non-functional targets

Attribute	Target	Mechanism
Availability SLO	99.9% monthly (43 min error budget)	Multi-AZ EKS node group, RDS Multi-AZ, HPA, PDB, multi-window burn-rate alerts
Latency SLO	p95 < 500 ms	`HighLatency` alert, API SLO dashboard
RPO	≤ 5 min (RDS PITR), ≤ 24 h (cross-cloud)	RDS automated backups and binlog, nightly S3 → GCS/ Azure copy
RTO	≤ 1 h regional, ≤ 4 h cross-cloud	Terraform re-apply, ArgoCD re-sync to AKS; see ../run-books/disaster-recovery.md
Patch latency	Critical CVEs < 72 h	Trivy gate, ECR scan-on-push, managed node AMI rollouts, Ansible `harden.yml`
Audit evidence	Continuous	CloudTrail/EKS audit logs, Azure diagnostics, Loki 90-day retention, nightly compli-ance scan artefacts